1 Problem setting

We consider the dynamical field model:

$$u_t = \mathcal{L}(p)u + \mathcal{Q}(u,u) + \mathcal{C}(u,u,u) + \epsilon^n \left(P\mathrm{e}^{\mathrm{i}\omega t} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega t}\right) + \epsilon^m \left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx}\right), \tag{1}$$

where $u(x,t) \in \mathbb{R}^d$, $x \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$, $p \in \mathbb{R}$, and $\mathcal{L}(p) = \mathcal{L}_0 + p\mathcal{L}_1 + O(p^2)$. The last two terms represent the weak forcing. We further assume the model satisfies:

1. $\mathcal{Q}$ and $\mathcal{C}$ are symmetric multilinear maps.
2. $\mathcal{L}_j$ is real, translation invariant, and reflection invariant, $j = 0, 1$.

Since the domain is periodic, every function of x is a linear combination of Fourier modes e^{ihx} where $h \in \mathbb{Z} \setminus \{0\}$. The operators $\mathcal{L}_j$, for $j = 0, 1$, are linear, hence for a spatial Fourier mode, define

$$\mathcal{L}_j\left(v\mathrm{e}^{\mathrm{i}hx}\right) = M_j(h)v\mathrm{e}^{\mathrm{i}hx}, \qquad M_j(h) \in \mathbb{C}^{d\times d}, \qquad h \in \mathbb{Z}\setminus\{0\}. \tag{2}$$

Due to reality and reflection symmetry, respectively,

$$M_j(-h) = \overline{M_j(h)}, \qquad M_j(-h) = M_j(h). \tag{3}$$

Thus $M_j(h)$ is real in this setting and the matrices at h and $-h$ are identical.

1.1 Bifurcation assumption

Now, further assume the Hopf bifurcation scenario: at $p = 0$, all eigenvalues of $M_0(h)$ are nonzero, except for one conjugate pair of eigenvalues for an unique spatial frequency $k_0 \in \mathbb{N}$, so that

$$M_0(\pm k_0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M_0(\pm k_0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \Psi^*\Phi = 1. \tag{4}$$

Here, $\Phi, \Psi \in \mathbb{C}^d$ are corresponding right and adjoint eigenvectors. Hence the only critical modes are the Hopf modes $(\Omega, h) = (\pm\omega_0, k_0), (\pm\omega_0, -k_0)$, where $\omega_0 > 0$ and $k_0 \in \mathbb{N}$. The conjugate equations give the eigenvectors $\overline{\Phi}$ and $\overline{\Psi}$ at eigenvalues $-\mathrm{i}\omega_0$ and $+\mathrm{i}\omega_0$ of M_0 and M_0^*, respectively. Because of (3), the same Φ, Ψ can be used at $\pm k_0$.

Let $\nu(h,p)$ be the eigenvalue branch through $\nu(\pm k_0, 0) = \mathrm{i}\omega_0$. If $M(\pm k_0, p)\Phi(p) = \nu(\pm k_0, p)\Phi(p)$, differentiation at $p = 0$ gives

$$M_1(\pm k_0)\Phi + M_0(\pm k_0)\Phi_p = \nu_p\Phi + \mathrm{i}\omega_0\Phi_p. \tag{5}$$

Since $\Psi^* M_0(\pm k_0) = \mathrm{i}\omega_0\Psi^*$, multiplication by Ψ^* cancels both terms containing Φ_p. Therefore

$$\partial_p\nu(\pm k_0, 0) = \Psi^* M_1(\pm k_0)\Phi =: \alpha, \qquad \operatorname{Re}\alpha > 0. \tag{6}$$

The last inequality is to set p as the bifurcation parameter. $\alpha \neq 0$ will only be used.

Furthermore, for $(\Omega, h) \notin \{(\pm\omega_0, \pm k_0)\}$, define the modal resolvent:

$$R(\Omega, h) := \left(\mathrm{i}\Omega I - M_0(h)\right)^{-1}. \tag{7}$$

Because the system is real, $M_0(-h) = \overline{M_0(h)}$, we have $R(-\Omega, -h) = \overline{R(\Omega, h)}$. In particular,

$$R(-\omega, 0) = \overline{R(\omega, 0)}, \qquad R(0, -k) = \overline{R(0, k)}. \tag{8}$$

With the reflection symmetry assumed here, $M_0(-k) = M_0(k)$ as well, so $R(0,-k) = R(0,k) = \overline{R(0,k)}$. Since $(\Omega, h) = (\pm\omega_0, k_0), (\pm\omega_0, -k_0)$ are only modes, we assume no quadratic resonance:

$$2\mathrm{i}\omega_0 I - M_0(0), \quad 2\mathrm{i}\omega_0 I - M_0(\pm 2k_0), -M_0(0), \quad -M_0(\pm 2k_0) \quad \text{are invertible.} \tag{9}$$

We additionally assume that $R(\pm\omega, 0)$ and $R(0, \pm k)$ exist when the corresponding forcing is present.

1.2 Scales

We introduce fast and slow times

$$s = t, \qquad \tau = \epsilon^2 t, \qquad \partial_t = \partial_s + \epsilon^2 \partial_\tau, \tag{10}$$

and initially write

$$p = \epsilon p_1 + \epsilon^2 p_2 + O(\epsilon^3), \qquad u = \epsilon u_1 + \epsilon^2 u_2 + \epsilon^3 u_3 + O(\epsilon^4). \tag{11}$$

The terms that determine the first three equations are

$$\begin{aligned}
\partial_t u = &\quad \epsilon\, \partial_s u_1 &&+\epsilon^2\, \partial_s u_2 &&+\epsilon^3 \big(\partial_s u_3 + \partial_\tau u_1\big) + O(\epsilon^4),\\
\mathcal{L}_0 u = &\quad \epsilon\, \mathcal{L}_0 u_1 &&+\epsilon^2\, \mathcal{L}_0 u_2 &&+\epsilon^3\, \mathcal{L}_0 u_3 + O(\epsilon^4),\\
p\mathcal{L}_1 u = &&&\epsilon^2 p_1 \mathcal{L}_1 u_1 &&+\epsilon^3 \big(p_2 \mathcal{L}_1 u_1 + p_1 \mathcal{L}_1 u_2\big) + O(\epsilon^4),\\
\mathcal{Q}(u,u) = &&&\epsilon^2 \mathcal{Q}(u_1,u_1) &&+2\epsilon^3 \mathcal{Q}(u_1,u_2) + O(\epsilon^4),\\
\mathcal{C}(u,u,u) = &&&&&\epsilon^3 \mathcal{C}(u_1,u_1,u_1) + O(\epsilon^4).
\end{aligned} \tag{12}$$

2 The solution space of $\mathcal{D} = \partial_s - \mathcal{L}_0$

First, let's define the linear operator and wave forms:

$$\mathcal{D} := \partial_s - \mathcal{L}_0, \qquad E_\pm := \mathrm{e}^{\mathrm{i}(\omega_0 s \pm k_0 x)}. \tag{13}$$

For any pure wave,

$$\mathcal{D}\left(v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (\mathrm{i}\Omega I - M_0(h))\, v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{14}$$

From the assumption, the multiplier in (14) is only singular at $(\Omega, h) = (\omega_0, \pm k_0)$ with unique nullvector Φ, or its conjugate which is singular at $(-\omega_0, \mp k_0)$ with nullvector $\overline{\Phi}$. Hence,

$$\ker \mathcal{D} = \mathrm{span}_{\mathbb{R}} \left\{ \Phi E_+, \Phi E_-, \overline{\Phi}\, \overline{E}_+, \overline{\Phi}\, \overline{E}_- \right\}. \tag{15}$$

2.1 Inner product

On functions periodic in x and in s with $T_0 = 2\pi/\omega_0$, use

$$\langle v, w \rangle := \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} v(s,x)^* w(s,x)\, dx\, ds. \tag{16}$$

Periodic integration by parts gives

$$\mathcal{D}^* = -\partial_s - \mathcal{L}_0^*, \qquad \mathcal{D}^*\left(Z\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (-\mathrm{i}\Omega I - M_0(h)^*)\, Z\mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{17}$$

So (4) gives the kernel space of $\mathcal{D}^*$:

$$\ker \mathcal{D}^* = \mathrm{span}_{\mathbb{R}} \left\{ \Psi E_+, \Psi E_-, \overline{\Psi}\, \overline{E}_+, \overline{\Psi}\, \overline{E}_- \right\}. \tag{18}$$

For waves that are Fourier basis,

$$\left\langle a\mathrm{e}^{\mathrm{i}(\Omega s + hx)}, b\mathrm{e}^{\mathrm{i}(\Omega' s + h' x)} \right\rangle = a^* b\, \delta_{\Omega,\Omega'} \delta_{h,h'}. \tag{19}$$

Thus different wave types are orthogonal. Having the same frequency and wavenumber pair is necessary for a nonzero inner product.

2.2 Fredholm alternative

For the notational convenience, let

$$U_\pm = \Phi E_\pm, \qquad V_\pm = \Psi E_\pm. \tag{20}$$

The complete pairings needed below are

	$\mathcal{L}_1 U_+$	$\mathcal{L}_1 U_-$	$\mathcal{L}_1 \overline{U}_+$	$\mathcal{L}_1 \overline{U}_-$
V_+	α	0	0	0
V_-	0	α	0	0
$\overline{V}_+$	0	0	$\overline{\alpha}$	0
$\overline{V}_-$	0	0	0	$\overline{\alpha}$

. (21)

Using the Fredholm alternative,

$$\operatorname{Ran}\mathcal{D} = (\ker \mathcal{D}^*)^\perp, \qquad \mathcal{D}w = f \iff \langle V, f\rangle = 0 \quad \text{for every } V \in \ker \mathcal{D}^*. \tag{22}$$

At every order, the goal is to put the right-hand side in $\operatorname{Ran}\mathcal{D}$.

2.3 Forcing scales

To prevent the stimulus itself from supplying the leading pattern, take $n, m > 1$. If $n = 1$ or $m = 1$, the corresponding stimulus appears in $\mathcal{D}u_1$ and u_1 is no longer chosen solely from $\ker\mathcal{D}$. If an exponent equals 2, its response is part of u_2 and interacts with u_1 through $2\mathcal{Q}(u_1, u_2)$ at order three. An exponent greater than 2 cannot produce such a cubic-order feedback. Thus, we only consider

$$F_2 = \mathbf{1}_{\{n=2\}}\left(P\mathrm{e}^{\mathrm{i}\omega s} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}}\left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx}\right). \tag{23}$$

The three equations at each scales are

$$O(\epsilon): \quad \mathcal{D}u_1 = 0, \tag{24a}$$

$$O(\epsilon^2): \quad \mathcal{D}u_2 = p_1\mathcal{L}_1 u_1 + \mathcal{Q}(u_1, u_1) + F_2, \tag{24b}$$

$$O(\epsilon^3): \quad \mathcal{D}u_3 = -\partial_\tau u_1 + p_2\mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2) + \mathcal{C}(u_1, u_1, u_1). \tag{24c}$$

3 $O(\epsilon)$ solution

From the equation

$$\mathcal{D}u_1 = 0,$$

the real leading solution is

$$\boxed{u_1 = A(\tau)\Phi E_+ + B(\tau)\Phi E_- + \text{c.c.}.} \tag{25}$$

4 $O(\epsilon^2)$ solution

The system at $O(\epsilon^2)$ level is:

$$\mathcal{D}u_2 = p_1\mathcal{L}_1 u_1 + \mathcal{Q}(u_1, u_1) + F_2.$$

There are wave types of $\{-2\omega_0, 0, 2\omega_0\} \times \{-2k_0, 0, 2k_0\}$ in the term $\mathcal{Q}(u_1, u_1)$ and $\{(\pm\omega, 0), (0, \pm k)\}$ in the term F_2. Hence, $p_1\mathcal{L}_1 u_1$ need to satisfy the Fredholm alternative.

4.1 Fredholm alternative check

The right-hand side of (24b) must be orthogonal to $\ker \mathcal{D}^*$. From the wave types of $\mathcal{Q}(u_1, u_1)$,

$$\langle V_\pm, \mathcal{Q}(u_1, u_1)\rangle = 0. \tag{26}$$

Similarly, for the forcing, $\langle V_\pm, F_2\rangle = 0$. Hence, only $p_1\mathcal{L}_1 u_1$ remains. By (21),

$$\langle V_+, p_1\mathcal{L}_1 u_1\rangle = p_1 A\alpha, \qquad \langle V_-, p_1\mathcal{L}_1 u_1\rangle = p_1 B\alpha. \tag{27}$$

Thus the two independent complex conditions are

$$0 = p_1 A\alpha, \qquad 0 = p_1 B\alpha. \tag{28}$$

For $(A, B) \neq (0, 0)$ and $\operatorname{Re}\alpha > 0$,

$$\boxed{p_1 = 0, \qquad p = \epsilon^2\mu + O(\epsilon^3).} \tag{29}$$

Hence, the system at $O(\epsilon^2)$ level is:

$$\mathcal{D}u_2 = \mathcal{Q}(u_1, u_1) + F_2.$$

4.2 Term $u_2^Q := \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1)$

Set the u_1 function as:

$$u_1^+ := \Phi(AE_+ + BE_-), \qquad u_1 = u_1^+ + \overline{u_1^+}. \tag{30}$$

Symmetry of $\mathcal{Q}$ gives

$$\mathcal{Q}(u_1, u_1) = \mathcal{Q}(u_1^+, u_1^+) + 2\mathcal{Q}(u_1^+, \overline{u_1^+}) + \mathcal{Q}(\overline{u_1^+}, \overline{u_1^+}). \tag{31}$$

Therefore, with $q_{20} := \mathcal{Q}(\Phi, \Phi)$ and $q_{11} := \mathcal{Q}(\Phi, \overline{\Phi})$,

$$\begin{aligned}\mathcal{Q}(u_1, u_1) = {} & q_{20}\left(A^2E_+^2 + B^2E_-^2 + 2ABE_+E_-\right) + \text{c.c.} \\ & + 2q_{11}\left(|A|^2 + |B|^2 + A\overline{B}E_+\overline{E}_- + \overline{A}B\overline{E}_+E_-\right).\end{aligned} \tag{32}$$

The generated wave types are

$$(\pm 2\omega_0, \pm 2k_0), \quad (\pm 2\omega_0, 0), \quad (0, \pm 2k_0), \quad (0, 0).$$

The condition (9) is the additional statement that $\mathcal{D}$ is invertible on each of them. Recall:

$$\mathcal{D}^{-1}\left(fe^{\mathrm{i}(\Omega s + hx)}\right) = R(\Omega, h)fe^{\mathrm{i}(\Omega s + hx)}.$$

Set $R\mathcal{Q}$ for each case as:

$$\begin{aligned} &h_{20}^+ = R(2\omega_0, 2k_0)q_{20}, && h_{20}^- = R(2\omega_0, -2k_0)q_{20}, && h_{20}^0 = 2R(2\omega_0, 0)q_{20}, \\ &h_{11}^0 = 2R(0, 0)q_{11}, && h_{11}^+ = 2R(0, 2k_0)q_{11}, && h_{11}^- = 2R(0, -2k_0)q_{11}. \end{aligned}$$

Reflection symmetry gives $h_{20}^+ = h_{20}^-$ and $h_{11}^+ = h_{11}^-$, but the superscripts are retained to display the wave types. Each terms in the quadratic term and it's effect on u_2 is:

$\mathcal{Q}(u_1, u_1)$	$A^2q_{20}E_+^2$	$B^2q_{20}E_-^2$	$2ABq_{20}E_+E_-$	$2(\lvert A\rvert^2 + \lvert B\rvert^2)q_{11}$	$2A\overline{B}q_{11}E_+\overline{E}_-$	$2\overline{A}Bq_{11}\overline{E}_+E_-$
(Ω, h)	$(2\omega_0, 2k_0)$	$(2\omega_0, -2k_0)$	$(2\omega_0, 0)$	$(0, 0)$	$(0, 2k_0)$	$(0, -2k_0)$
u_2^Q	$A^2h_{20}^+E_+^2$	$B^2h_{20}^-E_-^2$	$ABh_{20}^0E_+E_-$	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$	$A\overline{B}h_{11}^+E_+\overline{E}_-$	$\overline{A}Bh_{11}^-\overline{E}_+E_-$

Hence,

$$\begin{aligned}u_2^Q := \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1) = {} & A^2h_{20}^+E_+^2 + B^2h_{20}^-E_-^2 + ABh_{20}^0E_+E_- + \text{c.c.} \\ & + (|A|^2 + |B|^2)h_{11}^0 + A\overline{B}h_{11}^+E_+\overline{E}_- + \overline{A}Bh_{11}^-\overline{E}_+E_-.\end{aligned} \tag{33}$$

4.3 Term $u_2^F := \mathcal{D}^{-1}F_2$

For the the forcing frequency and wavenumber, define:

$$p_\omega := R(\omega, 0)P, \qquad s_k := R(0, k)S. \tag{34}$$

The symmetry (8) gives

$$R(-\omega, 0)\overline{P} = \overline{p_\omega}, \qquad R(0, -k)\overline{S} = \overline{s_k}. \tag{35}$$

By taking the inverse operator of $\mathcal{D}$, we get

$$u_2^F := \mathcal{D}^{-1}F_2 = \mathbf{1}_{\{n=2\}}\left(p_\omega \mathrm{e}^{\mathrm{i}\omega s} + \overline{p_\omega}\mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}}\left(s_k \mathrm{e}^{\mathrm{i}kx} + \overline{s_k}\mathrm{e}^{-\mathrm{i}kx}\right). \tag{36}$$

The conjugate terms make u_2^F real. The $O(\epsilon^2)$ solution therefore is:

$$\boxed{u_2 = u_2^Q + u_2^F.} \tag{37}$$

if there are no added components from $\ker \mathcal{D}$.

5 $O(\epsilon^3)$ solution and solvability

With $p_2 = \mu$, write the right-hand side of (24c) in two groups:

$$\mathcal{D}u_3 = \underbrace{-\partial_\tau u_1 + \mu\mathcal{L}_1 u_1 + 2\mathcal{Q}(u_1, u_2^Q) + \mathcal{C}(u_1, u_1, u_1)}_{=:F_0^{(3)}} + \underbrace{2\mathcal{Q}(u_1, u_2^F)}_{=:F_{Force}^{(3)}}. \tag{38}$$

In this section, we focus on the coefficients of the critical wave E_+, to analyze the solvability with Fredholm alternative. Note that for other critical waves, coefficients will be similar.

5.1 Unforced terms $F_0^{(3)}$

For the critical wave E_+, the linear term gives coefficients

$$[-\partial_\tau u_1 + \mu\mathcal{L}_1 u_1]_{E_+} = -A_\tau \Phi + \mu A M_1(k_0)\Phi. \tag{39}$$

For the cubic term, a triple of critical phases can sum to E_+ only as $E_+E_+\overline{E}_+$ or $E_+E_-\overline{E}_-$:

$$[\mathcal{C}(u_1, u_1, u_1)]_{E_+} = 3A|A|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}) + 6A|B|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}), \tag{40}$$

where 3 and 6 are the permutation counts. For the quadratic term, E_+ emerges at:

u_1 terms	$A\Phi E_+$	$\overline{A}\,\overline{\Phi}\,\overline{E}_+$	$\overline{B}\,\overline{\Phi}\,\overline{E}_-$	$B\Phi E_-$
u_2^Q terms	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$	$A^2 h_{20}^+ E_+^2$	$AB h_{20}^0 E_+ E_-$	$A\overline{B}h_{11}^+ E_+\overline{E}_-$
$2\mathcal{Q}$ terms	$2A(\lvert A\rvert^2 + \lvert B\rvert^2)\mathcal{Q}(\Phi, h_{11}^0)$	$2A\lvert A\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^+)$	$2A\lvert B\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^0)$	$2A\lvert B\rvert^2\mathcal{Q}(\Phi, h_{11}^+)$

No other pair of wave types sums to (ω_0, k_0). Summing the last column gives

$$[2\mathcal{Q}(u_1, u_2^Q)]_{E_+} = 2A(|A|^2 + |B|^2)\mathcal{Q}(\Phi, h_{11}^0) + 2A|A|^2\mathcal{Q}(\overline{\Phi}, h_{20}^+) \tag{41}$$

$$+ 2A|B|^2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2A|B|^2\mathcal{Q}(\Phi, h_{11}^+). \tag{42}$$

For the simplicity, define two coefficient terms for $A|A|^2$ and $A|B|^2$:

$$G_{\mathrm{s}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi}), \tag{43a}$$

$$G_{\mathrm{c}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2\mathcal{Q}(\Phi, h_{11}^+) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi}). \tag{43b}$$

With the same steps on E_-, we get similar coefficients with A and B are exchanged. Therefore

$$[F_0^{(3)}]_{E_+} = -A_\tau\Phi + \mu A M_1(k_0)\Phi + A|A|^2 G_{\mathrm{s}} + A|B|^2 G_{\mathrm{c}}, \tag{44}$$

$$[F_0^{(3)}]_{E_-} = -B_\tau\Phi + \mu B M_1(k_0)\Phi + B|B|^2 G_{\mathrm{s}} + B|A|^2 G_{\mathrm{c}}. \tag{45}$$

These and their conjugates are the only unforced terms outside $\operatorname{Ran}\mathcal{D}$.

5.2 Forcing-interaction terms $F_{Force}^{(3)}$

Recall that

$$u_1 = A\Phi E_+ + B\Phi E_- + \bar{A}\bar{\Phi}\bar{E_+} + \bar{B}\bar{\Phi}\bar{E_-},$$

and

$$u_2^F = u_{2,t}^F + u_{2,x}^F, \quad u_{2,t}^F \coloneqq \mathbf{1}_{\{n=2\}}\left(p_\omega \mathrm{e}^{\mathrm{i}\omega s} + \overline{p_\omega}\mathrm{e}^{-\mathrm{i}\omega s}\right), \quad u_{2,x}^F \coloneqq \mathbf{1}_{\{m=2\}}\left(s_k \mathrm{e}^{\mathrm{i}kx} + \overline{s_k}\mathrm{e}^{-\mathrm{i}kx}\right).$$

Hence, the wave types falls into the form of one of the followings:

$$\exp\left[\pm\mathrm{i}((\omega_0 \pm \omega)s \pm k_0 x)\right], \qquad \exp\left[\pm\mathrm{i}(\omega_0 s \pm (k_0 \pm k)x)\right],$$

where each $\pm$ are not correlated. To have nonzero critical wave terms, the only possible case is when $\omega = 0, \pm 2\omega_0$ and $k = 0, \pm 2k_0$. With $\omega, k \geq 0$, we have coefficients for the temporal forcing effects:

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_+} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2A\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \tag{46}$$

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_-} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2B\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right]. \tag{47}$$

In the same way, we have coefficients for the spatial forcing effects:

$$[2\mathcal{Q}(u_1, u_{2,x}^F)]_{E_+} = \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2A\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2B\mathcal{Q}(\Phi, s_{2k_0})\right], \tag{48}$$

$$[2\mathcal{Q}(u_1, u_{2,x}^F)]_{E_-} = \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2B\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2A\mathcal{Q}(\Phi, \overline{s_{2k_0}})\right]. \tag{49}$$

5.3 Fredholm conditions

Using $\Psi^*\Phi = 1$, the projection to the kernel space of $\mathcal{D}^*$ is:

$$\left\langle V_+, F^{(3)}\right\rangle = \Psi^*[F^{(3)}]_{E_+}, \qquad \left\langle V_-, F^{(3)}\right\rangle = \Psi^*[F^{(3)}]_{E_-}. \tag{50}$$

Define values

$$a \coloneqq \mu\Psi^* M_1(k_0)\Phi = \mu\alpha, \qquad b \coloneqq \Psi^* G_{\mathrm{s}}, \qquad c \coloneqq \Psi^* G_{\mathrm{c}}, \tag{51}$$

and denote forcing projections as:

$$\begin{aligned}
\mathcal{T}_A &= \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2A\Psi^*\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \\
\mathcal{T}_B &= \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2B\Psi^*\mathcal{Q}(\Phi, p_0 + \overline{p_0}) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})\right], \\
\mathcal{X}_A &= \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2A\Psi^*\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2B\Psi^*\mathcal{Q}(\Phi, s_{2k_0})\right], \\
\mathcal{X}_B &= \mathbf{1}_{\{m=2\}}\left[\mathbf{1}_{\{k=0\}} 2B\Psi^*\mathcal{Q}(\Phi, s_0 + \overline{s_0}) + \mathbf{1}_{\{k=2k_0\}} 2A\Psi^*\mathcal{Q}(\Phi, \overline{s_{2k_0}})\right].
\end{aligned}$$

Then, the Fredholm alternative now gives

$$0 = -A_\tau + A(a + b|A|^2 + c|B|^2) + \mathcal{T}_A + \mathcal{X}_A, \tag{52a}$$

$$0 = -B_\tau + B(a + b|B|^2 + c|A|^2) + \mathcal{T}_B + \mathcal{X}_B. \tag{52b}$$

6 Amplitude equations and patterns

Considering only non-homogeneous forcing at $2\omega_0$ and $2k_0$, define the complex forcing coefficients

$$\eta_t := 2\Psi^* \mathcal{Q}\big(\overline{\Phi}, p_{2\omega_0}\big)\,, \qquad \eta_+ := 2\Psi^* \mathcal{Q}(\Phi, s_{2k_0})\,, \qquad \eta_- := 2\Psi^* \mathcal{Q}(\Phi, \overline{s_{2k_0}})\,.$$

Thus $\eta_t = 0$ when $P = 0$, and $\eta_x = 0$ when $S = 0$. The amplitude equations are

$$A_\tau = A\left(a + b|A|^2 + c|B|^2\right) + \eta_t \overline{B} + \eta_+ B, \tag{53a}$$
$$B_\tau = B\left(a + b|B|^2 + c|A|^2\right) + \eta_t \overline{A} + \eta_- A. \tag{53b}$$

Here $a, b, c, \eta_t, \eta_x \in \mathbb{C}$. An equilibrium satisfies

$$0 = A\left(a + b|A|^2 + c|B|^2\right) + \eta_t \overline{B} + \eta_+ B, \tag{54}$$
$$0 = B\left(a + b|B|^2 + c|A|^2\right) + \eta_t \overline{A} + \eta_- A. \tag{55}$$

6.1 Existence of a nonzero pattern below onset

Take $n = m = 2$, $\omega = 2\omega_0$, $k = 2k_0$, and

$$p = \epsilon^2 \mu, \qquad \mu < 0, \qquad a = \mu\alpha. \tag{56}$$

All scaled coefficients are fixed as $\epsilon \to 0$. Assume $S \in \mathbb{R}^d$, so that s_{2k_0} is real and $\eta_+ = \eta_- =: \eta_x$. Set

$$a_+ = a + \eta_x, \qquad \gamma = b + c. \tag{57}$$

Since $\operatorname{Re} a = \mu \operatorname{Re} \alpha < 0$, the unforced zero-amplitude solution is linearly stable.

Proposition 1. *If $\gamma \neq 0$ and $|\eta_t| > |a_+|$, then (53) has a nonzero equilibrium $A_* = B_* = z_*$. Its leading field is a standing pattern of order ϵ.*

Proof. The subspace $A = B = z$ is invariant, and its equation is

$$z_\tau = z(a_+ + \gamma|z|^2) + \eta_t \overline{z}. \tag{58}$$

For $z = \sqrt{\rho}\, \mathrm{e}^{\mathrm{i}\phi} \neq 0$, the equilibrium condition gives

$$a_+ + \gamma\rho + \eta_t \mathrm{e}^{-2\mathrm{i}\phi} = 0, \qquad \rho > 0. \tag{59}$$

Taking absolute values from $a_+ + \gamma\rho = -\eta_t \mathrm{e}^{-2\mathrm{i}\phi}$, define

$$f(\rho) := |a_+ + \gamma\rho|^2 - |\eta_t|^2 = |\gamma|^2\rho^2 + 2\operatorname{Re}(a_+\overline{\gamma})\rho + |a_+|^2 - |\eta_t|^2. \tag{60}$$

Its leading coefficient is positive and its constant term is negative. Hence it has exactly one positive root ρ_*, with $f'(\rho_*) > 0$. Choose ϕ_* by

$$\mathrm{e}^{-2\mathrm{i}\phi_*} = -\frac{a_+ + \gamma\rho_*}{\eta_t}, \qquad z_* = \sqrt{\rho_*}\, \mathrm{e}^{\mathrm{i}\phi_*}. \tag{61}$$

The quotient has modulus one, so $A_* = B_* = z_*$ is an equilibrium. The leading field is therefore

$$u(x,t) = 4\epsilon\sqrt{\rho_*}\, \operatorname{Re}\left[\Phi \mathrm{e}^{\mathrm{i}(\omega_0 t + \phi_*)}\right] \cos(k_0 x) + O(\epsilon^2). \tag{62}$$

Since $\rho_* > 0$, $\Phi \neq 0$, and $k_0 \neq 0$, its leading term is nonzero and spatially nonconstant. □

Stability. At $z = 0$, (58) has eigenvalues

$$\lambda_\pm = \operatorname{Re} a_+ \pm \sqrt{|\eta_t|^2 - (\operatorname{Im} a_+)^2}. \tag{63}$$

The condition $|\eta_t| > |a_+|$ gives $\lambda_+ > 0$. Thus the forcing destabilizes zero critical amplitude.

At $(A_*, B_*) = (z_*, z_*)$, write

$$v_\pm = \tfrac{1}{2}\mathrm{e}^{-\mathrm{i}\phi_*}(\delta A \pm \delta B). \tag{64}$$

Using (59), the linearized equations separate as

$$(v_+)_\tau = (a_+ + 2\gamma\rho_*)v_+ - a_+\overline{v}_+, \tag{65}$$

$$(v_-)_\tau = (a - \eta_x + 2b\rho_*)v_- + (a_+ + 2b\rho_*)\overline{v}_-. \tag{66}$$

For $v_\tau = hv + \ell\overline{v}$, the real Jacobian has trace $2\operatorname{Re} h$ and determinant $|h|^2 - |\ell|^2$. Thus

$$\begin{aligned} \operatorname{tr} J_+ &= 2\operatorname{Re}(a_+ + 2\gamma\rho_*), & \det J_+ &= 2\rho_* f'(\rho_*) > 0, \\ \operatorname{tr} J_- &= 2\operatorname{Re}(a - \eta_x + 2b\rho_*), & \det J_- &= -4\operatorname{Re}[(a + 2b\rho_*)\overline{\eta_x}]\,. \end{aligned} \tag{67}$$

Consequently the equilibrium is strictly linearly stable precisely when

$$\operatorname{Re}(a_+ + 2\gamma\rho_*) < 0, \quad \operatorname{Re}(a - \eta_x + 2b\rho_*) < 0, \quad \operatorname{Re}[(a + 2b\rho_*)\overline{\eta_x}] < 0. \tag{68}$$

For pure temporal forcing, $S = 0$ and $\eta_x = 0$. In this case J_- has eigenvalues 0 and $2\operatorname{Re}(a+2b\rho_*)$. The zero eigenvalue is tangent to the translation family

$$(A, B) = (z_*\mathrm{e}^{\mathrm{i}\theta}, z_*\mathrm{e}^{-\mathrm{i}\theta}), \qquad \theta \in \mathbb{R}. \tag{69}$$

Since $\operatorname{Re} a < 0$, the sufficient conditions

$$\boxed{|\eta_t| > |\mu\alpha|, \qquad \operatorname{Re} b < 0, \qquad \operatorname{Re}(b + c) < 0} \tag{70}$$

make the other three eigenvalues have negative real part. The standing wave is therefore locally asymptotically stable modulo spatial translations in the amplitude system.

6.2 Notation summary

Table 1. Shorthand quantities in Sections 1–6.

Symbol	Definition	Location
$M_j(h)$	Fourier matrix: $\mathcal{L}_j(v\mathrm{e}^{\mathrm{i}hx}) = M_j(h)v\mathrm{e}^{\mathrm{i}hx}$.	(2)
Φ, Ψ	Critical right and adjoint eigenvectors; $\Psi^*\Phi = 1$.	(4)
α	$\Psi^* M_1(k_0)\Phi = \partial_p \nu(k_0, 0)$.	(6)
$\mathcal{D}, R(\Omega, h)$	$\mathcal{D} = \partial_s - \mathcal{L}_0$, $R(\Omega, h) = (\mathrm{i}\Omega I - M_0(h))^{-1}$.	(13), (7)
$E_\pm, U_\pm, V_\pm$	$E_\pm = \mathrm{e}^{\mathrm{i}(\omega_0 s \pm k_0 x)}$, $U_\pm = \Phi E_\pm$, $V_\pm = \Psi E_\pm$.	Section 2
A, B	$u_1 = A\Phi E_+ + B\Phi E_- + \text{c.c.}$.	(25)
q_{20}, q_{11}	$q_{20} = \mathcal{Q}(\Phi, \Phi)$, $q_{11} = \mathcal{Q}(\Phi, \overline{\Phi})$.	Section 4.2
$h_{20}^\pm, h_{20}^0$	$h_{20}^\pm = R(2\omega_0, \pm 2k_0)q_{20}$, $h_{20}^0 = 2R(2\omega_0, 0)q_{20}$.	Section 4.2
$h_{11}^\pm, h_{11}^0$	$h_{11}^\pm = 2R(0, \pm 2k_0)q_{11}$, $h_{11}^0 = 2R(0, 0)q_{11}$.	Section 4.2
p_ω, s_k	$p_\omega = R(\omega, 0)P$, $s_k = R(0, k)S$.	(34)
u_2^Q, u_2^F	$u_2^Q = \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1)$, $u_2^F = \mathcal{D}^{-1}F_2$.	(33), (36)
G_s	$2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi})$.	(43a)
G_c	$2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2\mathcal{Q}(\Phi, h_{11}^+) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi})$.	(43b)
a, b, c	$a = \mu\alpha$, $b = \Psi^* G_\mathrm{s}$, $c = \Psi^* G_\mathrm{c}$, with $\mu = p_2$.	(51)
$\mathcal{T}_{A,B}, \mathcal{X}_{A,B}$	Temporal and spatial forcing projections onto $V_\pm$; upper sign for A, lower sign for B.	Section 5.3
$\eta_t, \eta_\pm$	$\eta_t = 2\Psi^*\mathcal{Q}(\overline{\Phi}, p_{2\omega_0})$, $\eta_+ = 2\Psi^*\mathcal{Q}(\Phi, s_{2k_0})$, $\eta_- = 2\Psi^*\mathcal{Q}(\Phi, \overline{s_{2k_0}})$.	Section 6
η_x, a_+, γ	$\eta_x = \eta_+ = \eta_-$ for real S; $a_+ = a + \eta_x$, $\gamma = b + c$.	(57)
$z, \rho, f(\rho)$	$A = B = z$, $\rho = \lvert z\rvert^2$, $f(\rho) = \lvert a_+ + \gamma\rho\rvert^2 - \lvert\eta_t\rvert^2$.	(58)–(60)
ρ_*, ϕ_*, z_*	$f(\rho_*) = 0$, $\rho_* > 0$, $z_* = \sqrt{\rho_*}\mathrm{e}^{\mathrm{i}\phi_*}$; $\mathrm{e}^{-2\mathrm{i}\phi_*} = -(a_+ + \gamma\rho_*)/\eta_t$.	(61)
$v_\pm, J_\pm$	$v_\pm = \frac{1}{2}\mathrm{e}^{-\mathrm{i}\phi_*}(\delta A \pm \delta B)$; $J_\pm$ are their real 2×2 linearization matrices.	(65)–(67)

7 V1-inspired model and parameter choice

In this section we consider 4 types of populations: (E, PV, SST, VIP). For the connectivity matrix, rows index targets and columns index sources. Pfeffer et al. [1] motivate the retained inhibitory motif. Exact zeros are a reduction assumption. Billeh et al. [2] supply the cell-pair distance scales. Campagnola et al. [3] document the dependence of effective connectivity on cell class and layer.

Units, activation, and domain. Time is in ms, u, F, I are dimensionless, and

$$\ell_x = 100\,\mu\text{m}, \qquad L = 43, \qquad L_{\text{phys}} = 4.3\,\text{mm}, \qquad \tau = (10, 12, 35, 20)\,\text{ms}. \tag{71}$$

Retain the effective activation parameters

$$\begin{aligned} F_j(u) &= \ell(g_j(u-\theta_j)) - \ell(-g_j\theta_j), \qquad \ell(v) = (1+\mathrm{e}^{-v})^{-1}, \\ g &= (3.47682761, 7.34413013, 3.74650527, 6), \\ \theta &= (.28483799, .12068148, .40137356, .7). \end{aligned} \tag{72}$$

Literature-based widths and fixed gains. Billeh et al. [2], STAR Methods, p. e11, give $s_{E\to E} = 114$, $s_{E\to PV} = 92$, $s_{E\to SST} = 103$, $s_{PV\to E} = 95$, and $s_{SST\to E} = 85\,\mu$m, based on Levy and Reyes (2012), and $s_{PV\to PV} \simeq 120\,\mu$m from internal Allen Institute data. They use the PV–PV scale for other inhibitory pairs and the E–SST scales for E–Htr3a pairs. Replacing Htr3a by VIP is our additional reduction. Their distance rule is $\exp(-r^2/s_{ij}^2)$. Hence s_{ij} is the exponential distance scale and the standard deviation in our convention is $s_{ij}/\sqrt{2}$. With target rows and source columns,

$$s = \begin{pmatrix} 114 & 95 & 85 & - \\ 92 & 120 & 120 & - \\ 103 & - & - & 120 \\ 103 & - & 120 & - \end{pmatrix} \mu\text{m}, \quad \Sigma = \begin{pmatrix} .6498 & .45125 & .36125 & - \\ .4232 & .72 & .72 & - \\ .53045 & - & - & .72 \\ .53045 & - & .72 & - \end{pmatrix}, \quad \Sigma_{ij} = \frac{s_{ij}^2}{2\ell_x^2}. \tag{73}$$

Here Σ is the variance in model coordinates and $\Sigma^{\text{phys}} = \ell_x^2\Sigma$, with $\ell_x = 100\,\mu$m. A dash denotes an absent connection ($G_{ij} = 0$). For the effective integrated gains, we use

$$G = \begin{pmatrix} 6.22009 & -1.45455 & -6.48944 & 0 \\ 5.09798 & -.90615 & -8.04066 & 0 \\ 2.67174 & 0 & 0 & -16 \\ 1.68844 & 0 & -3.13584 & 0 \end{pmatrix}. \tag{74}$$

These magnitudes are fixed model coefficients. They are not measured connection probabilities.

Increase the scalar κ in $J = \kappa G$ from zero, compute its first modal crossing κ_c, and set

$$p = \kappa - \kappa_c, \qquad J(p) = (\kappa_c + p)G, \qquad J_0 = \kappa_c G, \quad J_1 = G. \tag{75}$$

The periodic-domain computation gives

$$\kappa_c = .7491489065, \qquad k_0 = 3, \qquad q_0 = \frac{6\pi}{43}, \qquad \omega_0 = .0829864799\,\text{rad/ms}. \tag{76}$$

Thus $f_0 = 13.2077085\,$Hz and the wavelength is $4.3/3$ mm.

Input populations. For visual/thalamic input, take $e = (1, 1, 0, 0)^{\mathsf{T}}$. Direct E and PV innervation is supported by Ji et al. [4] and used in Billeh et al.'s feedforward model. Equal normalized strengths are our assumption, not a measured ratio.

7.1 Computation of the amplitude coefficients

All quantities below are evaluated at onset $p = 0$, with $q_0 = 6\pi/43$ and ω_0 given in (76). With dimensionless amplitudes and $\tau = \epsilon^2 t$ measured in ms, $\alpha, a, b, c, c_\perp$ and all forcing coefficients η have units ms^{-1}. Matrices M, M_p and the multilinear maps Q_q, C_q likewise have units ms^{-1} on dimensionless arguments; $\mathcal{R}$ has units ms, and the response vectors h, $p_{2\omega_0}$, s_{2q_0} are dimensionless.

1. Activation derivatives and Fourier maps. Set $r_j = \ell(-g_j\theta_j)$. Direct differentiation gives

$$\begin{aligned} F_j'(0) &= g_j r_j(1-r_j), \\ F_j''(0) &= g_j^2 r_j(1-r_j)(1-2r_j), \\ F_j'''(0) &= g_j^3 r_j(1-r_j)(1-6r_j+6r_j^2). \end{aligned} \tag{77}$$

Form $D_F = \mathrm{diag}(F_j'(0))$ and $\boldsymbol{F}_r = (F_j^{(r)}(0))_j$, $r = 2, 3$. Substitute $\widehat{W}(q;0) = \kappa_c G \odot \mathrm{e}^{-\Sigma q^2/2}$ into (89) and (91) to obtain $M(q,0)$, Q_q, and C_q.

2. Critical eigenvectors and linear coefficient. Solve

$$M(q_0,0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M(q_0,0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \|\Phi\|_2 = 1, \quad \Psi^*\Phi = 1. \tag{78}$$

Then

$$\alpha = \Psi^* D_\tau^{-1}(G \odot \mathrm{e}^{-\Sigma q_0^2/2}) D_F \Phi, \qquad a = \mu\alpha. \tag{79}$$

The unit-norm convention fixes the amplitude scale of b, c.

3. Quadratic responses. With $\mathcal{R}(\Omega, q) = (\mathrm{i}\Omega I_K - M(q,0))^{-1}$, solve

$$\begin{aligned} h_{20}^+ &= \mathcal{R}(2\omega_0, 2q_0) Q_{2q_0}(\Phi,\Phi), & h_{20}^0 &= 2\mathcal{R}(2\omega_0, 0) Q_0(\Phi,\Phi), \\ h_{11}^0 &= 2\mathcal{R}(0,0) Q_0(\Phi,\overline{\Phi}), & h_{11}^+ &= 2\mathcal{R}(0, 2q_0) Q_{2q_0}(\Phi,\overline{\Phi}). \end{aligned} \tag{80}$$

Each Q_q uses the output wavenumber of that interaction. Reflection symmetry gives the corresponding negative-wavevector responses.

4. Cubic coefficients. Project the resonant terms at output wavenumber q_0:

$$\begin{aligned} b &= \Psi^* \left[2Q_{q_0}(\Phi, h_{11}^0) + 2Q_{q_0}(\overline{\Phi}, h_{20}^+) + 3C_{q_0}(\Phi,\Phi,\overline{\Phi}) \right], \\ c &= \Psi^* \left[2Q_{q_0}(\Phi, h_{11}^0) + 2Q_{q_0}(\overline{\Phi}, h_{20}^0) + 2Q_{q_0}(\Phi, h_{11}^+) + 6C_{q_0}(\Phi,\Phi,\overline{\Phi}) \right]. \end{aligned} \tag{81}$$

The computed values, in ms^{-1}, are

$$\alpha = .0880577 + .0643807\mathrm{i}, \qquad b = -6.706479 + 9.069454\mathrm{i}, \qquad c = -6.961796 + 8.999612\mathrm{i}. \tag{82}$$

5. Forcing coefficients and input strength. For an input $I = \epsilon^2(P\mathrm{e}^{2\mathrm{i}\omega_0 t} + S\mathrm{e}^{2\mathrm{i}q_0 x} + \mathrm{c.c.})$ in (86), compute

$$p_{2\omega_0} = \mathcal{R}(2\omega_0, 0) D_\tau^{-1} P, \qquad s_{2q_0} = \mathcal{R}(0, 2q_0) D_\tau^{-1} S \tag{83}$$

$$\eta_t = 2\Psi^* Q_{q_0}(\overline{\Phi}, p_{2\omega_0}), \quad \eta_+ = 2\Psi^* Q_{q_0}(\Phi, s_{2q_0}), \quad \eta_- = 2\Psi^* Q_{q_0}(\Phi, \overline{s_{2q_0}}). \tag{84}$$

Here P, S are dimensionless input harmonics. The factors D_τ^{-1} convert them to rates in ms^{-1} in the u_t equation. For real S, $\eta_+ = \eta_- =: \eta_x$.

Let the default E/PV drive $e = (1, 1, 0, 0)^\mathsf{T}$. Take $\mu = -1$, $\epsilon = .02$, hence $p = -.0004$. The forcing coefficients are

$$\eta_t^{(1)} = .055509 - .029075\mathrm{i}, \quad \eta_x^{(1)} = .018902 + .046191\mathrm{i} \quad (\mathrm{ms}^{-1}). \tag{85}$$

8 Dispersion matrix and frequency variation

8.1 Fourier matrix for K populations

On $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$, consider the source-nonlinearity field

$$D_\tau u_t = -u + W(p) * F(u) + I(x,t), \qquad D_\tau = \mathrm{diag}(\tau_1, \dots, \tau_K), \quad \tau_i > 0, \tag{86}$$

where $F(u) = (F_j(u_j))_{j=1}^K$ is the activation function, $F(0) = 0$, and rows of W index targets, columns index sources. Use periodized, unit-mass Gaussian kernels of variances $\Sigma_{ij} > 0$:

$$W_{ij}(x;p) = J_{ij}(p) \sum_{\ell \in \mathbb{Z}} \frac{\mathrm{e}^{-(x+L\ell)^2/(2\Sigma_{ij})}}{\sqrt{2\pi\Sigma_{ij}}}, \qquad J(p) = J_0 + pJ_1. \tag{87}$$

Their convolution multipliers, with $q_h = 2\pi h/L$, are

$$\widehat{W}_{ij}(q;p) = J_{ij}(p)\mathrm{e}^{-\Sigma_{ij}q^2/2}, \qquad q = |q_h|, \quad h \in \mathbb{Z}. \tag{88}$$

Here $\widehat{W}$ is the integral Fourier transform of the kernel on the torus, without the factor L^{-1}.

Set $D_F = \mathrm{diag}(F_1'(0), \dots, F_K'(0))$. Linearizing at $u = 0$, $I = 0$ and substituting $u = v\mathrm{e}^{\nu t + \mathrm{i}q_h x}$ gives

$$\nu D_\tau v = -v + \widehat{W}(q;p) D_F v, \qquad \boxed{M(q,p) = D_\tau^{-1}\left[-I_K + \widehat{W}(q;p)D_F\right].} \tag{89}$$

Thus the dispersion relation is

$$\mathcal{P}(\nu, q, p) := \det(\nu I_K - M(q,p)) = 0. \tag{90}$$

The parameter p is measured from onset, as in the preceding sections. For $L = 2\pi$, $M_0(h) = M(|h|, 0)$ and $M_1(h) = \partial_p M(|h|, 0)$.

For this field the quadratic and cubic maps act on Fourier modes and depend on the output wavenumber q. With $\boldsymbol{F}_r = (F_j^{(r)}(0))_{j=1}^K$,

$$Q_q(v,w) = \tfrac{1}{2}D_\tau^{-1}\widehat{W}(q;0)(\boldsymbol{F}_2 \odot v \odot w), \qquad C_q(v,w,z) = \tfrac{1}{6}D_\tau^{-1}\widehat{W}(q;0)(\boldsymbol{F}_3 \odot v \odot w \odot z), \tag{91}$$

where $\odot$ is the entrywise product. In Sections 2–5 and 9–10, $\mathcal{Q}$ and $\mathcal{C}$ stand for Q_q and C_q evaluated at the wavenumber of the output wave type, and $R(\Omega, q) = \mathcal{R}(\Omega, q)D_\tau^{-1}$ acts on inputs.

Two-population example. The E–I Wilson–Cowan field of the PI's code is (86) with $K = 2$ and

$$\begin{gathered} J = \begin{pmatrix} \kappa & -10 \\ 10 & -8 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 8 & 32 \\ 8 & 32 \end{pmatrix}, \\ \tau = (3, 6.6), \quad g = (4,4), \quad \theta = (.5, .3), \quad L = 60, \quad e = (1,0)^{\mathsf{T}}. \end{gathered} \tag{92}$$

Here $\kappa = a_{EE}$, the kernels are $\mathrm{e}^{-(x/\sigma)^2}$ with $\sigma_E = 4, \sigma_I = 8$, and time is in model units. The onset is $a_{EE}^* = 7.024920$, $k_0 = 3$, $\omega_0 = .318128$, with

$$\begin{gathered} \alpha = .047165 - .048825\mathrm{i}, \qquad b = -.715856 - .545818\mathrm{i}, \qquad c = -1.248695 + 4.101512\mathrm{i}, \\ \eta_t^{(1)} = .287550 + .034009\mathrm{i}, \qquad \eta_x^{(1)} = 1.049746 + .491768\mathrm{i}. \end{gathered} \tag{93}$$

Since $\mathrm{Re}\, c < \mathrm{Re}\, b$, the unforced pattern is a traveling wave. On the square torus with $L = 60$ the oblique shell $|\boldsymbol{k}|^2 = 10$ loses stability first, at $a_{EE} = 7.022974$; Section 10 chooses L so that one shell is critical.

8.2 Eigenvalue derivatives

Fix p. Let $\nu(q)$ be a simple eigenvalue with right and left eigenvectors

$$Mv = \nu v, \qquad w^*M = \nu w^*, \qquad w^*v = 1. \tag{94}$$

Differentiation gives $M_q v + Mv_q = \nu_q v + \nu v_q$. Multiplying by w^* cancels the terms containing v_q, hence

$$\boxed{\nu_q = w^*M_q v, \qquad \nu_p = w^*M_p v,} \tag{95}$$

where, with entrywise exponentiation,

$$\begin{aligned} M_q &= -qD_\tau^{-1}\left[(\Sigma \odot \widehat{W}(q;p))D_F\right], \\ M_p &= D_\tau^{-1}\left[(J_1 \odot \mathrm{e}^{-\Sigma q^2/2})D_F\right]. \end{aligned} \tag{96}$$

The second identity assumes that F and D_τ do not depend on p. At the critical point, $v = \Phi$, $w = \Psi$, and $\nu_p = \alpha$.

Write $\nu = \sigma + \mathrm{i}\omega$ with $\omega > 0$. Then

$$\sigma_q = \mathrm{Re}(w^*M_q v), \qquad \boxed{\omega_q = \mathrm{Im}(w^*M_q v).} \tag{97}$$

Equivalently, since $\mathcal{P}(\nu(q), q, p) = 0$ holds identically along the branch, implicit differentiation of (90) gives

$$\nu_q = -\frac{\mathcal{P}_q}{\mathcal{P}_\nu}, \qquad \boxed{\omega_q = \frac{\mathrm{Im}(\mathcal{P}_\nu \overline{\mathcal{P}_q})}{|\mathcal{P}_\nu|^2},} \tag{98}$$

with all derivatives of $\mathcal{P}$ evaluated at $(\nu(q), q, p)$. Because ν is a simple root, $\mathcal{P}_\nu \neq 0$, so the sign of ω_q is the sign of the numerator $\mathrm{Im}(\mathcal{P}_\nu \overline{\mathcal{P}_q})$.

8.3 The rightmost eigenvalue

The formulas above follow a single eigenvalue branch. The frequency that matters for the instability is that of the eigenvalue with the largest real part. We therefore define the spectral abscissa $\sigma_{\max}$ and the frequency $\omega_{\mathrm{r}} \geq 0$ of the rightmost eigenvalue by

$$\sigma_{\max}(q) = \max_{\nu \in \mathrm{spec}\, M(q,p)} \mathrm{Re}\,\nu, \qquad \omega_{\mathrm{r}}(q) = \max\left\{\mathrm{Im}\,\nu : \mathrm{Re}\,\nu = \sigma_{\max}(q)\right\}. \tag{99}$$

Since M is real, complex eigenvalues come in conjugate pairs. If a pair is rightmost, ω_{r} is its positive imaginary part. If a unique real eigenvalue is rightmost, $\omega_{\mathrm{r}} = 0$.

V1 parameters. For the parameters of Section 7, at $p = 0$ and $q_0 = 6\pi/43$,

$$\omega_q(q_0) = .0471593 > 0, \qquad \frac{d\omega}{dq_{\mathrm{phys}}} = .00471593\,\mathrm{mm/ms}, \tag{100}$$

where q is the model-coordinate wavenumber and ω is in rad/ms. The rightmost frequencies of modes $k = 0, 1, 2, 3, 4$ are 11.0865, 11.3759, 12.1588, 13.2077, and 14.2833 Hz, so the frequency increases across these modes; this does not imply global monotonicity. Changing L changes which q values are sampled, but not the derivative at fixed q and κ.

8.4 Shape of the frequency curve

In simulations the frequency $\omega(q)$ of the oscillatory pair is bell shaped for V1, but it looks monotone over the modes that matter. This subsection explains both observations.

The frequency must come back to zero. Every multiplier decays, $\widehat{W}(q;p) \to 0$ as $q \to \infty$. Hence

$$M(q,p) \to -D_\tau^{-1} \qquad (q \to \infty), \tag{101}$$

which is real and diagonal. Its eigenvalues $-1/\tau_i$ are real, so every imaginary part tends to zero. If the τ_i are distinct, the limit eigenvalues are simple and real, so all eigenvalues are real for q large enough. Hence a positive frequency at $q = 0$ cannot increase for all q: it must eventually return to zero. This rules out a globally increasing branch. It does not fix the shape in between. In both models below the curve has one maximum (observed numerically): at $q = 0$ (decreasing curve) or at $q > 0$ (a bell).

An increase needs heterogeneity. If all $\tau_i = \tau_c$ and all retained $\Sigma_{ij} = \Sigma_c$, then $M = (-I + \mathrm{e}^{-\Sigma_c q^2/2} J D_F)/\tau_c$ and every eigenvalue is $\nu(q) = (-1 + \mathrm{e}^{-\Sigma_c q^2/2}\zeta)/\tau_c$ for some fixed $\zeta \in \mathrm{spec}(JD_F)$. Thus

$$\omega_q = -\Sigma_c\, q\, \omega < 0 \qquad (q > 0), \tag{102}$$

and the frequency decreases everywhere. A rising branch, as in (100), requires unequal time constants or unequal widths.

The two models. Figure 1 follows the pair at onset. For the E–I Wilson–Cowan model the bell is centred at $q = 0$: the frequency only falls, from 0.904 at $q = 0$ to $\omega_0 = 0.318$ at q_0, and the pair becomes real at $2.16q_0$. For V1 the frequency rises from 0.0697 to a maximum 0.1033 at $q \simeq 2.61q_0$ and then falls. The pair stops being the rightmost eigenvalue at $q \simeq 2.54q_0$, just before the maximum. On the modes where the pair sets the instability, ω_r therefore increases monotonically. The three-population model (the E/PV/SST block of G with G_{EE} raised from 6.22009 to 6.72, so that $k_0 = 3$ again) has its maximum at $2.67q_0$ and stays rightmost up to $3.79q_0$, so there the full bell is visible.

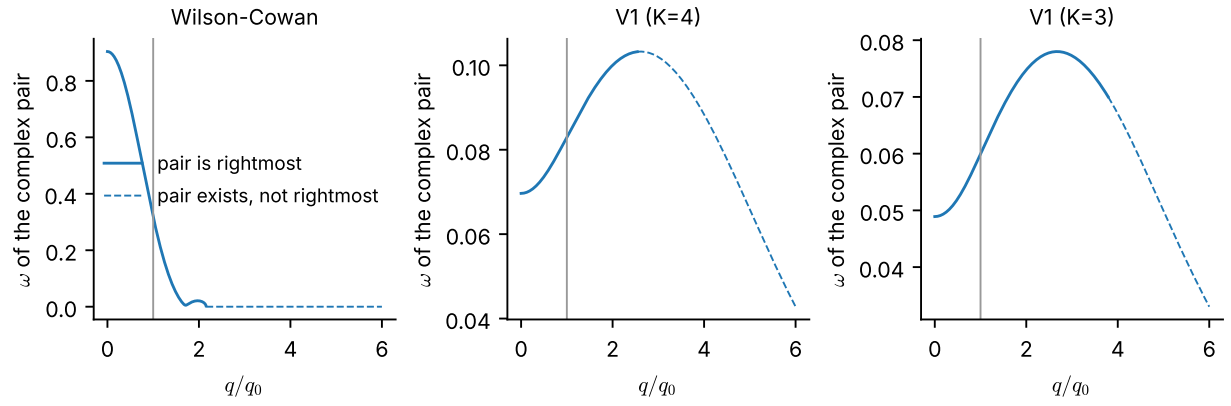


Figure 1: Frequency of the complex pair at onset versus q/q_0. Solid: the pair is the rightmost eigenvalue. Dashed: the pair exists but another eigenvalue is to its right. The vertical line marks q_0.

Why the shape matters. A uniform input at frequency Ω is resonant with the modes that satisfy $2\omega(q) = \Omega$. On a monotone branch there is one such q, so a faster flicker selects a finer (V1) or a coarser (Wilson–Cowan) pattern. On a bell there can be two such q, and none above $2\max\omega$. This is an observation about the linear resonance condition only; on the torus only the lattice values $q = 2\pi h/L$ are available.

9 Forcing strength and wavenumber

Sections 5–7 used the resonant inputs $2\omega_0$ and $2k_0$ at order ϵ^2. Here we ask what happens for a static input at any other wavenumber k, and how weak it may be. The answer follows from one counting rule.

9.1 Counting rule

Take the static input $\epsilon^m(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx})$, $k \in \mathbb{N}$, and $p = \epsilon^2\mu$. Its response is $s_k\mathrm{e}^{\mathrm{i}kx} + \text{c.c.}$ with $s_k = R(0,k)S$, as in (34). This response exists for every k because the critical modes oscillate, $\omega_0 \neq 0$. The input changes the amplitude equations only through terms that are resonant with E_+. With r factors of the input and one critical wave, the wave types are

r	wave types	resonant with E_+ when
1	$(\omega_0,\ \pm k_0 \pm k)$	$k = 0$ or $k = 2k_0$
2	$(\omega_0,\ \pm k_0 \pm k \pm k)$	always $(k - k = 0)$, and $k = k_0$

A term with r factors has order ϵ^{rm+1}. It must meet $a\,A$ at order ϵ^3. Hence

$$\boxed{k = 2k_0:\ m = 2,\ \ \text{input} \propto \epsilon^2; \qquad k \notin \{0, 2k_0\}:\ m = 1,\ \ \text{input} \propto \epsilon.} \tag{103}$$

Section 2.3 excluded $m = 1$ to keep u_1 in $\ker\mathcal{D}$. For a non-resonant input, the extra term $s_k\mathrm{e}^{\mathrm{i}kx}$ in u_1 is a forced response, not a free amplitude, so $m = 1$ is admissible. The same counting for a uniform input at frequency ω gives $\omega \in \{0, 2\omega_0\}$ with one factor, and $\omega = \omega_0$ or any ω with two factors (unchecked: we have not computed these coefficients). The table stops at $r = 2$, that is, at order ϵ^3. With r factors, $rk = 2k_0$ is resonant at a higher order.

9.2 Amplitude equations for $m = 1$

Let $k \notin \{0, 2k_0\}$ and S real. At order ϵ^3 the input contributes $A\,\chi$ for every k, and $B\,\xi$ only for $k = k_0$:

$$A_\tau = A\left(a + \chi + b|A|^2 + c|B|^2\right) + \xi B, \tag{104a}$$

$$B_\tau = B\left(a + \chi + b|B|^2 + c|A|^2\right) + \xi A. \tag{104b}$$

Here $\mathcal{C}$ acts pointwise, as in the field (86). The coefficients collect the second-order responses, as in (43):

$$\begin{aligned}\chi = \Psi^*\Big[&2\mathcal{Q}\big(\Phi, 2R(0,0)\mathcal{Q}(s_k, \overline{s_k})\big) + 4\mathcal{Q}\big(\overline{s_k}, R(\omega_0, k_0 + k)\mathcal{Q}(\Phi, s_k)\big)\\ &+ 4\mathcal{Q}\big(s_k, R(\omega_0, k_0 - k)\mathcal{Q}(\Phi, \overline{s_k})\big) + 6\mathcal{C}(\Phi, s_k, \overline{s_k})\Big],\\ \xi = \mathbf{1}_{\{k=k_0\}}\Psi^*\Big[&2\mathcal{Q}\big(\Phi, R(0, 2k_0)\mathcal{Q}(s_k, s_k)\big) + 4\mathcal{Q}\big(s_k, R(\omega_0, 0)\mathcal{Q}(\Phi, s_k)\big) + 3\mathcal{C}(\Phi, s_k, s_k)\Big].\end{aligned} \tag{105}$$

Both are quadratic in the input. At $A = B = 0$ the combinations $A \pm B$ decouple with growth rates $\mathrm{Re}(\mu\alpha + \chi \pm \xi)$. For $k = 2k_0$, Section 6 gives $\mathrm{Re}(\mu\alpha \pm \eta_x)$ instead. Write the input as a cosine of peak δ. With $\mathrm{Re}\,\alpha > 0$ (6), the threshold moves to

$$\boxed{p_{\mathrm{th}} = -\frac{\mathrm{Re}\,\chi^{(1)} + |\,\mathrm{Re}\,\xi^{(1)}|}{\mathrm{Re}\,\alpha}\,\delta^2 \quad (k \notin \{0, 2k_0\}), \qquad p_{\mathrm{th}} = -\frac{|\,\mathrm{Re}\,\eta_x^{(1)}|}{\mathrm{Re}\,\alpha}\,|\delta| \quad (k = 2k_0),} \tag{106}$$

where the superscript (1) marks the value for $\delta = 1$. At this order only $k = k_0$ and $k = 2k_0$ lock the pattern to the input. Every other k moves the threshold but leaves the spatial phase free.

9.3 The two-parameter diagram

Figure 2 repeats the (δ, a_{EE}) diagram for the E–I Wilson–Cowan model on $L = 60$ ($k_0 = 3$). For each stimulus wavenumber k it shows the a_{EE} at which the stimulus-locked steady state first loses stability. Near $\delta = 0$, (106) explains the two shapes.

- $k = 6 = 2k_0$: the two branches $A = \pm B$ are straight lines with slopes ± 22.26 that cross at $\delta = 0$. The threshold is a V. A continuation that follows one Hopf branch through $\delta = 0$ traces one line across the other, which gives the loop.
- $k \le 3$: both branches are parabolas tangent at $\delta = 0$, so the curve is smooth there.

The quadratic coefficients and a check against the full model at small q are (shifts in units of 10^{-4})

k	1	2	3 ($= k_0$)	4	5	6 ($= 2k_0$)
coefficient of δ^2 or $\lvert\delta\rvert$	3.38	5.32	13.9	662	2032	−22.26
shift, full model	0.84	1.32	3.48	26.6	82.2	−437
shift, theory	0.84	1.33	3.48	26.5	81.3	−445

with $\delta = 0.005$ for $k \le 3$ and $\delta = 0.002$ otherwise. Near $\delta = 0$, every k except $2k_0$ raises the threshold. The coefficients for $m = 4, 5$ are large because $R(0, k)$ is large there. These modes have eigenvalues close to zero, $-0.025 + 0.115\mathrm{i}$ and $-0.084 + 0.009\mathrm{i}$, which is the falling frequency curve of Section 8.4. Their quadratic range is therefore small, $|\delta| \lesssim 0.002$. Far from $\delta = 0$ all curves bend down, outside the weakly nonlinear range.

10 Two dimensions

On the square torus $\mathbb{T}^2_L$ the critical modes form a shell: all lattice wavevectors $\boldsymbol{k}$ with $|\boldsymbol{k}| = q_0$. Each gives a wave $E_{\boldsymbol{k}} = \mathrm{e}^{\mathrm{i}(\omega_0 s + \boldsymbol{k}\cdot\boldsymbol{x})}$ with amplitude $A_{\boldsymbol{k}}$, and $u_1 = \sum_{\boldsymbol{k}} A_{\boldsymbol{k}} \Phi E_{\boldsymbol{k}} + \mathrm{c.c.}$ The 1D amplitudes are $A = A_{k_0\hat{x}}$ and $B = A_{-k_0\hat{x}}$. We use three assumptions.

Assumption 1. *M_0, $\mathcal{Q}$ and $\mathcal{C}$ depend on a wavevector only through its length (true for the field (86) with isotropic Gaussian kernels), and $\mathcal{C}$ acts pointwise.*

Assumption 2. *At $p = 0$ only the shell $|\boldsymbol{k}| = q_0$ has eigenvalues on the imaginary axis, the pair $\pm\mathrm{i}\omega_0$; all other lattice modes are strictly stable.*

Assumption 3. *For all shell waves, $2\mathrm{i}\omega_0 \notin \mathrm{spec}\, M_0(|\boldsymbol{k}_X + \boldsymbol{k}_Z|)$ and $0 \notin \mathrm{spec}\, M_0(|\boldsymbol{k}_X - \boldsymbol{k}_Y|)$ unless the difference is zero; for an input $\boldsymbol{K}$ at $\Omega = 2\omega_0$, also $2\mathrm{i}\omega_0 \notin \mathrm{spec}\, M_0(|\boldsymbol{K}|)$.*

Assumption 3 replaces (9). Under these assumptions the calculation of Sections 3–5 carries over word for word; only the bookkeeping of wavevectors changes.

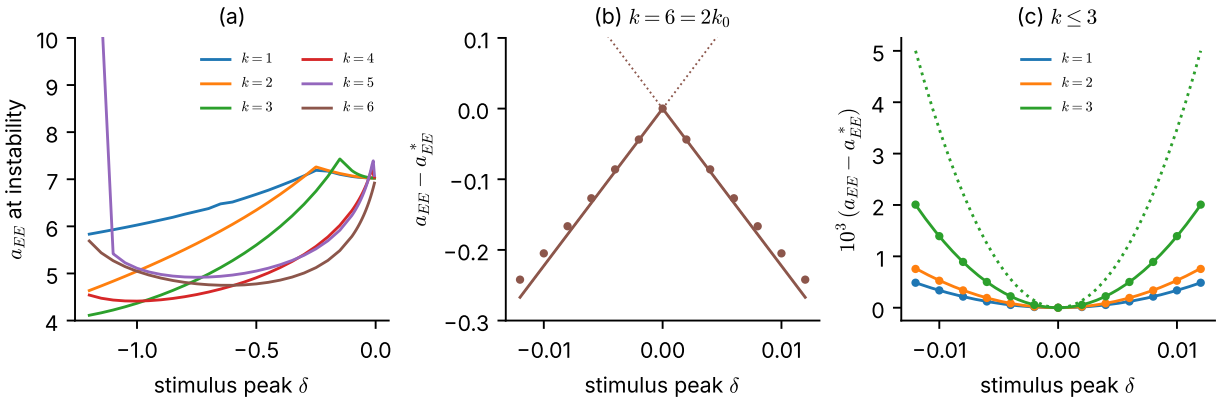


Figure 2: Static stimulus $\delta\cos(2\pi kx/60)$ on E, Wilson–Cowan, $N = 60$. (a) a_{EE} at the first instability of the stimulus-locked state. (b), (c) the same near $\delta = 0$: dots from the full model, lines from (106), dotted lines for the second branch.

10.1 Which waves an input couples

Take an input $\epsilon^2(H\mathrm{e}^{\mathrm{i}(\Omega t+\boldsymbol{K}\cdot\boldsymbol{x})} + \text{c.c.})$. At order ϵ^3 it couples two critical waves when their wave types add up, exactly as in (46)–(49):

input	coupled waves	term in $(A_{\boldsymbol{k}_Y})_\tau$
$\Omega = 2\omega_0$	$\boldsymbol{k}_X + \boldsymbol{k}_Y = \boldsymbol{K}$	$\eta(\lvert\boldsymbol{K}\rvert)\,\overline{A_{\boldsymbol{k}_X}}, \quad \eta(K) = 2\Psi^*\mathcal{Q}\big(\overline{\Phi}, R(2\omega_0, K)H\big)$
$\Omega = 0$	$\boldsymbol{k}_Y - \boldsymbol{k}_X = \pm\boldsymbol{K}$	$\eta_s^{\pm}\,A_{\boldsymbol{k}_X}, \quad \eta_s^+ = 2\Psi^*\mathcal{Q}\big(\Phi, R(0,K)H\big), \quad \eta_s^- = 2\Psi^*\mathcal{Q}\big(\Phi, R(0,K)\overline{H}\big)$

(107)

Here $K = |\boldsymbol{K}|$; the $-\boldsymbol{K}$ row comes from the complex conjugate of the input. For $\boldsymbol{K} = 0$ and $\Omega = 2\omega_0$ this is η_t; for $K = 2k_0$ and $\Omega = 0$ it gives $\eta_\pm$.

Geometry. Both coupled waves have length q_0. For $\Omega = 2\omega_0$ they are therefore placed symmetrically about $\boldsymbol{K}$:

$$\boldsymbol{k}_{X,Y} = \tfrac{1}{2}\boldsymbol{K} \pm \boldsymbol{d}, \qquad \boldsymbol{d} \perp \boldsymbol{K}, \qquad |\boldsymbol{d}|^2 = q_0^2 - \tfrac{1}{4}|\boldsymbol{K}|^2. \tag{108}$$

There are three cases.

- $\boldsymbol{K} = 0$ (uniform flicker): each wave couples to its opposite. No direction is preferred.
- $|\boldsymbol{K}| = 2q_0$: $\boldsymbol{d} = 0$ and the wave couples to itself. A traveling input selects one traveling wave along $\boldsymbol{K}$.
- $0 < |\boldsymbol{K}| < 2q_0$: the pair has a component $\pm\boldsymbol{d}$ across $\boldsymbol{K}$. An input traveling along x then creates structure along y.

On the torus, $\boldsymbol{k}_{X,Y}$ must be lattice vectors. On the axial shell $\{(\pm k_0, 0), (0, \pm k_0)\}$, an input along an axis can only have $K \in \{0, 2q_0\}$, so it produces no structure across itself; the diagonal input $\boldsymbol{K} = (k_0, k_0)$ couples $(k_0, 0)$ with $(0, k_0)$. On the oblique shell $\{(\pm 3, \pm 1), (\pm 1, \pm 3)\}$ the axial input $\boldsymbol{K} = (6, 0)$ couples $(3, 1)$ with $(3, -1)$.

10.2 Cubic terms

The cubic terms are found as in Section 5. The equation of $A_{\boldsymbol{k}}$ contains every monomial $A_{\boldsymbol{k}_X}\overline{A_{\boldsymbol{k}_Y}}A_{\boldsymbol{k}_Z}$ with $\boldsymbol{k}_X - \boldsymbol{k}_Y + \boldsymbol{k}_Z = \boldsymbol{k}$. For distinct $\boldsymbol{k}_X \neq \boldsymbol{k}_Z$, its coefficient is

$$\boxed{\Psi^*\Big[2\mathcal{Q}\big(\Phi, h_{11}(\boldsymbol{k}_X - \boldsymbol{k}_Y)\big) + 2\mathcal{Q}\big(\overline{\Phi}, h_{20}(\boldsymbol{k}_X + \boldsymbol{k}_Z)\big) + 2\mathcal{Q}\big(\Phi, h_{11}(\boldsymbol{k}_Z - \boldsymbol{k}_Y)\big) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi})\Big],} \tag{109}$$

with $h_{20}(\boldsymbol{v}) = 2R(2\omega_0, |\boldsymbol{v}|)\mathcal{Q}(\Phi, \Phi)$ and $h_{11}(\boldsymbol{v}) = 2R(0, |\boldsymbol{v}|)\mathcal{Q}(\Phi, \overline{\Phi})$. Each term is one pair of waves forming a second-order response, then meeting the third wave. For $\boldsymbol{k}_X = \boldsymbol{k}_Z$ (the monomial $A_{\boldsymbol{k}_X}^2\overline{A_{\boldsymbol{k}_Y}}$) each multiplicity halves:

$$\Psi^*\Big[\mathcal{Q}\big(\overline{\Phi}, h_{20}(2\boldsymbol{k}_X)\big) + 2\mathcal{Q}\big(\Phi, h_{11}(\boldsymbol{k}_X - \boldsymbol{k}_Y)\big) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi})\Big], \tag{110}$$

and $\boldsymbol{k}_X = \boldsymbol{k}_Y = \boldsymbol{k}_Z$ gives b (43a). Two further cases recur.

- Two waves at angle $\theta \neq 0$ ($\boldsymbol{k}_Y = \boldsymbol{k}_Z$): the cross coefficient c_θ, whose responses sit at $|\boldsymbol{k}_X \pm \boldsymbol{k}_Y| = 2q_0|\cos\frac{\theta}{2}|$ and $2q_0|\sin\frac{\theta}{2}|$. It gives $c_\pi = c$ and $c_{\pi/2} = c_\perp$.
- Four waves of the axial shell: the quartet term $c_\times \overline{A_{-\hat{x}}}A_{\hat{y}}A_{-\hat{y}}$ in the $A_{\hat{x}}$ equation (lattice directions in units of k_0).

10.3 The locked pair

Proposition 2. *Assume 1–3. Let an input with $\Omega = 2\omega_0$ couple the pair $\boldsymbol{k}_X \neq \boldsymbol{k}_Y$ of (108), at angle θ, and assume that no shell wave equals $2\boldsymbol{k}_X - \boldsymbol{k}_Y$ or $2\boldsymbol{k}_Y - \boldsymbol{k}_X$. Then $A_{\boldsymbol{k}_X} = A_{\boldsymbol{k}_Y} = z$ with all other amplitudes zero is invariant, and*

$$z_\tau = z\big(a + (b + c_\theta)|z|^2\big) + \eta\,\overline{z}. \tag{111}$$

This is (58) with $a_+ = a$ and $\gamma = b + c_\theta$. If $b + c_\theta \neq 0$ and $|\eta| > |\mu\alpha|$, Proposition 1 gives $z_* = \sqrt{\rho_*}\mathrm{e}^{\mathrm{i}\phi_*}$ and the leading field

$$\boxed{u = 4\epsilon\sqrt{\rho_*}\,\mathrm{Re}\Big[\Phi\mathrm{e}^{\mathrm{i}(\omega_0 t + \frac{1}{2}\boldsymbol{K}\cdot\boldsymbol{x} + \phi_*)}\Big]\cos(\boldsymbol{d}\cdot\boldsymbol{x}) + O(\epsilon^2).} \tag{112}$$

Proof. On the subspace, the monomials that contain only $A_{\boldsymbol{k}_X}$ and $A_{\boldsymbol{k}_Y}$ land on $\boldsymbol{k}_X$, $\boldsymbol{k}_Y$, $2\boldsymbol{k}_X - \boldsymbol{k}_Y$, or $2\boldsymbol{k}_Y - \boldsymbol{k}_X$. By assumption the last two are not shell waves, so all other amplitudes stay zero. An input term $\eta\overline{A_{\boldsymbol{k}_{X'}}}$ is nonzero on the subspace only if $\boldsymbol{k}_{X'} \in \{\boldsymbol{k}_X, \boldsymbol{k}_Y\}$, and then it lands on $\boldsymbol{K} - \boldsymbol{k}_{X'} \in \{\boldsymbol{k}_Y, \boldsymbol{k}_X\}$. The $A_{\boldsymbol{k}_X}$ equation contains $b|A_{\boldsymbol{k}_X}|^2$, $c_\theta|A_{\boldsymbol{k}_Y}|^2$, and, by (107), $\eta\overline{A_{\boldsymbol{k}_Y}}$. The $A_{\boldsymbol{k}_Y}$ equation is the same with X and Y exchanged, which gives (111). Finally $\mathrm{e}^{\mathrm{i}\boldsymbol{k}_X\cdot\boldsymbol{x}} + \mathrm{e}^{\mathrm{i}\boldsymbol{k}_Y\cdot\boldsymbol{x}} = 2\mathrm{e}^{\mathrm{i}\boldsymbol{K}\cdot\boldsymbol{x}/2}\cos(\boldsymbol{d}\cdot\boldsymbol{x})$. □

For $\boldsymbol{K} = 0$ this is the 1D standing pattern (62); the perpendicular pair is also driven but stays zero on the subspace. For $0 < |\boldsymbol{K}| < 2q_0$ the pattern travels with the input along $\boldsymbol{K}$ and is standing across it, with wavenumber $|\boldsymbol{d}| = (q_0^2 - |\boldsymbol{K}|^2/4)^{1/2}$ set by the input. An input traveling along x thus creates a spatial pattern along y. A static input ($\Omega = 0$) couples $\boldsymbol{k}_Y = \boldsymbol{k}_X + \boldsymbol{K}$ instead, which gives a pattern standing along $\boldsymbol{K}$ and traveling across it; we do not treat it further.

10.4 Stability inside the shell

The pair is stable in the whole problem only if the other shell waves decay. Their linearized growth involves c_θ at the angles to $\boldsymbol{k}_X$ and $\boldsymbol{k}_Y$, and the four-wave terms of (109). For uniform flicker on the axial shell this is explicit. At the stripe $A_{\pm\hat{x}} = z_*$, the perpendicular pair obeys $v_\tau = hv + \ell\,\overline{w}$ with

$$h = a + 2c_\perp\rho_*, \qquad \ell = \eta_t + c_\times\rho_* e^{2i\phi_*}, \qquad \boxed{\lambda_\perp = \mathrm{Re}\Big[h + \sqrt{|\ell|^2 - (\mathrm{Im}\,h)^2}\Big],} \tag{113}$$

so the stripe is stable across x when $\mathrm{Re}\,h < 0$ and $|\ell| < |h|$. The square $A_{\pm\hat{x}} = A_{\pm\hat{y}} = z$ obeys (111) with $b + c_\theta$ replaced by $\gamma_\square = b + c + 2c_\perp + c_\times$. For other shells we evaluate the full Jacobian numerically.

The sign of $\mathrm{Re}\,c_\theta$ decides the outcome. In V1 it is negative at every angle. In the E–I model it is positive for $71° \lesssim \theta \lesssim 146°$ (Figure 3). There, one wave amplifies another. A pattern destabilizes when the shell holds waves at such angles to it, and the cubic equations then cannot saturate. The sharp peak near 141° is a near resonance. There the sum of the two wavevectors has length $0.67q_0$, where M has eigenvalues $-0.039 \pm 0.566\mathrm{i}$, close to $2\omega_0 = 0.593$, so $R(2\omega_0, \cdot)$ is large. This is the falling frequency curve of Section 8.4 again: in the E–I model a longer wave oscillates faster and can match $2\omega_0$.

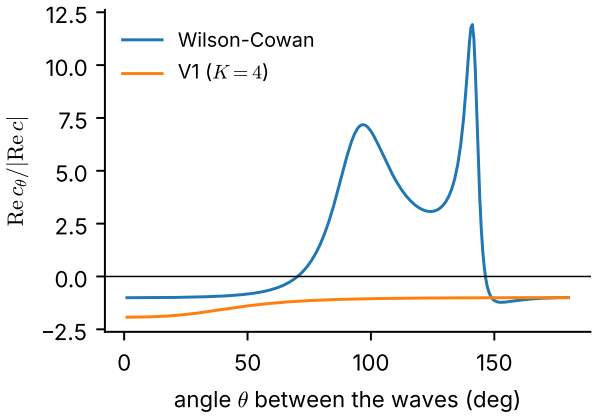


Figure 3: $\mathrm{Re}\,c_\theta$ against the angle between two critical waves, scaled by $|\mathrm{Re}\,c|$. Wilson–Cowan on $L = 58.26$ (axial shell critical) and V1 on $L = 43$.

10.5 Numerical tests

We integrate the full 2D field below onset ($\mu = -1$, 32×32 grid) and measure the amplitudes of the shell waves. The domain length is chosen so that the intended shell is critical: $L = 58.26$ (axial)

and $L = 61.42$ (oblique) for Wilson–Cowan, and $L = 43$ and $L = 45.01$ for V1.

test	model, ϵ	direct field	theory
y-wave growth $\mathrm{Re}(\mu\alpha + c_\perp\rho_T)$	WC, 0.1	$+0.588$	$+0.557$
	V1, 0.05 / 0.025	-0.0080 / -0.0098	-0.0103
flicker: stripe amplitude	WC, 0.1	0.02389	0.02385
	V1, 0.05 / 0.025	0.00341 / 0.00196	0.00409 / 0.00204
flicker: stripe $\lambda_\perp$	WC, 0.1	$+1.77$	$+1.84$
	V1, 0.05 / 0.025	-0.017 / -0.024	-0.026
flicker: square amplitude	V1, 0.05 / 0.025	0.00273 / 0.001373	0.00276 / 0.001378
x-traveling input: pair $\lvert A\rvert$	WC, 0.1	0.0264 (then leaves)	0.0270 (unstable)
	V1, 0.05 / 0.025	0.00397 / 0.00242	0.00509 / 0.00254 (stable)

The first test runs above onset ($\mu = +1$) on the traveling wave $A_{\hat{x}}$ of size $\rho_T = -\operatorname{Re}\alpha/\operatorname{Re}b$; rates are per slow time. The V1 flicker rows come from `theory/code/check_square_2d.py` (output in `results/`). In every V1 test the gap shrinks by a factor of 2.8 to 4.7 when ϵ is halved. In V1 the stripe under flicker is stable across x, while the square is unstable to a symmetry-breaking mode ($+0.022$). In the E–I model both are unstable: the stripe across x, and the square inside its own subspace ($+0.236$). Under the x-traveling input on the oblique shell, the pair $(3, \pm 1)$ is selected from noise in both models. In V1 the pair persists, and the other six waves stay small (below 2×10^{-6} at the end of the runs of Section 10.6). In the E–I model the largest eigenvalue of the shell Jacobian, $+0.84$, has its eigenvectors on the waves $(-1, \pm 3)$. We remove the $\mathrm{e}^{-2\mathrm{i}\omega_0 t}$ part of each $A_{\boldsymbol{k}}$ by a sliding fit, because the pair leaks into it. When the pair locks, this slow part of every other wave is at most 2.5×10^{-5}. After the lock, the waves $(-1, \pm 3)$ grow at 0.70–0.71 per slow time in three seeds, fitted while their slow part is between 10^{-5} and 10^{-4}. This is about 16% below the eigenvalue (observed; the Jacobian is taken at the theoretical z_*, with $\epsilon = 0.1$). The other shell waves grow with them, and the field leaves the weakly nonlinear range.

10.6 Random initial states

Most tests above start near the predicted state; the oblique test starts from small noise ($10^{-3}\epsilon$). In the two 2D notebooks (folder `notebooks`) every run starts instead from a random field of size 0.1ϵ (V1: $\epsilon = 0.025$; E–I: $\epsilon = 0.1$).

- x-traveling input, oblique shell $|\boldsymbol{k}|^2 = 10$, $\boldsymbol{K} = (6, 0)$: in V1 all three seeds select $(3, \pm 1)$, with $\epsilon|A| = 0.00242$ in each (theory 0.00254); every other shell wave stays below 2×10^{-6}. Each seed has its own y-position, as the translation mode predicts, and the pattern moves along x with the input.

- The same input on the shell $|\boldsymbol{k}|^2 = 25$ ($L = 71.17$) selects $(3, \pm 4)$: a pattern with y-mode 4 from an input with no y-structure. V1 gives 0.00271 in all three seeds (theory 0.00289; shell eigenvalues at most -0.263 apart from translation).

- Uniform flicker, axial shell: six V1 seeds all end as stripes, three varying along x and three along y; none ends as a square. The stripe amplitude $\sqrt{|A_+|^2 + |A_-|^2} = 0.00277$ (theory 0.00288).

- In the E–I model both inputs first produce the predicted pattern, which then leaves the weakly nonlinear range, as Section 10.5 predicts.

An independent rerun (seed 7, and a 48×48 grid) reproduced the oblique V1 result to three digits. Two cautions remain. For the x-traveling pair in V1 the gap to the theory is 22% at $\epsilon = 0.05$ and

5% at $\epsilon = 0.025$ (table of Section 10.5). The gap falls by a factor of about 4.7 when ϵ is halved, but only two values of ϵ were tested. At $\epsilon = 0.025$ the neighbouring shell $|\boldsymbol{k}|^2 = 9$ decays almost as slowly as the critical shell at $u = 0$: $\max \mathrm{Re}\, \nu = -7.3 \times 10^{-5}$ against -5.5×10^{-5} per ms ($|\boldsymbol{k}|^2 = 13$: -2.0×10^{-4}). So the single-shell reduction is close to its limit. In the three V1 runs the shells $|\boldsymbol{k}|^2 = 9, 13$ carry only the decaying initial noise. In the runs shell 9 decays faster than at $u = 0$, at -1.6 to -1.7×10^{-4} per ms (observed; we did not separate the effect of the pair from that of the input). The largest population Fourier coefficient on these shells is below 3×10^{-6} after 40 s and at most 3.4×10^{-9} at 80 s. For the pair it is 1.6×10^{-3}, which corresponds to $\epsilon|A| = 0.00242$ (`theory/code/check_shell_growth_2d.py`).

10.7 Summary

The 2D picture has three steps. The input wavevector selects which critical waves are coupled (107). The coupled pair is a 1D problem, and Proposition 1 gives its amplitude (111). The cross coefficients c_θ then decide whether that pair survives the rest of the shell. The multi-population V1 model passes the third step. The two-population E–I model with these parameters fails it: its perpendicular coupling is destabilizing.

10.8 Notation for Sections 8.4–10

Symbol	Definition	Location
s_k, δ	static response $R(0,k)S$ to an order-ϵ input; cosine peak of the input	Sec. 9
χ, ξ	input-induced shift and A–B coupling for $m = 1$	(105)
$E_{\boldsymbol{k}}, A_{\boldsymbol{k}}$	critical wave $\mathrm{e}^{\mathrm{i}(\omega_0 t + \boldsymbol{k}\cdot\boldsymbol{x})}$ and its amplitude, $\lvert\boldsymbol{k}\rvert = q_0$	Sec. 10
$\boldsymbol{K}, \eta(K), \eta_s^{\pm}$	input wavevector; coupling of a pair at $\Omega = 2\omega_0$ or $\Omega = 0$	(107)
$\boldsymbol{d}$	half difference of the coupled pair, $\perp \boldsymbol{K}$	(108)
$c_\theta, c_\perp, c_\times$	cross coefficient at angle θ; $c_\perp = c_{\pi/2}$; axial quartet coefficient	(109)
$\gamma_\square, \lambda_\perp$	square coefficient $b + c + 2c_\perp + c_\times$; transverse growth of the stripe	(113)

References

[1] C. K. Pfeffer, M. Xue, M. He, Z. J. Huang, and M. Scanziani. Inhibition of inhibition in visual cortex: the logic of connections between molecularly distinct interneurons. *Nature Neuroscience* 16 (2013), 1068–1076. doi:10.1038/nn.3446.

[2] Y. N. Billeh et al. Systematic Integration of Structural and Functional Data into Multi-scale Models of Mouse Primary Visual Cortex. *Neuron* 106 (2020), 388–403.e18; Fig. 4 and STAR Methods. doi:10.1016/j.neuron.2020.01.040.

[3] L. Campagnola et al. Local connectivity and synaptic dynamics in mouse and human neocortex. *Science* 375 (2022), eabj5861. doi:10.1126/science.abj5861.

[4] X.-Y. Ji et al. Thalamocortical Innervation Pattern in Mouse Auditory and Visual Cortex: Laminar and Cell-Type Specificity. *Cerebral Cortex* 26 (2016), 2612–2625. doi:10.1093/cercor/bhv099.